

Multiple biological functions of sporamin related to stress tolerance in sweet potato (*Ipomoea batatas* Lam).



The initial investigation of the nature of the proteins in the tuber of sweet potato (*Ipomoea batatas* Lam.) revealed a globulin-designated "ipomoein," which was reported by Jones and Gersdorff, (1931). Later, "ipomoein" was renamed "sporamin" and was found to be a major storage protein that accounted for over 80% of the total protein in the tuberous root. To date, sporamin has been studied by a series of biochemical and molecular approaches. The first purification of sporamin into two major fractions, A and B, was successfully completed in 1985. Several characteristics of the protein, such as the diversification of the nucleotide sequences in the gene family, the protein structure, the biological functions of storage, defense, inhibitory activity and ROS scavenging, were identified. In the past decade, sporamin was classified as a Kunitz-type trypsin inhibitor, and its insect-resistance capability has been examined in transgenic tobacco and cauliflower plants, indicating the multiple functions of this protein has evolved to facilitate the growth and development of sweet potato. Sporamin is constitutively expressed in the tuberous root and is not normally expressed in the stem or leaves. However, this protein is expressed systemically in response to wounding and other abiotic stresses. These dual expression patterns at the transcriptional level revealed that the complex regulatory mechanism of sporamin was modulated by environmental stresses. The versatile functions of sporamin make this storage protein a good research model to study molecular evolution, regulatory mechanisms and physiological functions in plants. This review summarizes and discusses recent approaches and future perspectives in agricultural biotechnology. Copyright © 2012 Elsevier Inc. All rights reserved.